

SAFETY DATA SHEET



Date Prepared : 02/01/2016
SDS No : NDT-06-595-A

ADDITIVE 595

1. PRODUCT AND COMPANY IDENTIFICATION

PRODUCT NAME: ADDITIVE 595
PRODUCT DESCRIPTION: Booster Detergent for Acidic Cleaners

SUPPLIER

Broadmoor Products, Inc.
4201 Brockton Drive, S.E.
Grand Rapids, MI 49512
Customer Service: (616) 285-6440

24 HR. EMERGENCY TELEPHONE NUMBER

P.E.R.S. (US Transportation/ Spill): (800) 633-8253
For after-hours transportation and spill emergencies only.
Direct all other calls to the Customer Service number at left.

2. HAZARDS IDENTIFICATION

GHS CLASSIFICATIONS

Health:

Serious Eye Damage, Category 1
Skin Irritation, Category 2

Environmental:

Acute Hazards to the Aquatic Environment, Category 2
Chronic Hazards to the Aquatic Environment, Category 2

GHS LABEL



Corrosion



Environment

SIGNAL WORD: DANGER

HAZARD STATEMENTS

H318: Causes serious eye damage.
H315: Causes skin irritation.
H411: Toxic to aquatic life with long lasting effects.

PRECAUTIONARY STATEMENT(S)

Prevention:

P280: Wear protective gloves/protective clothing/eye protection/face protection when working with the concentrated product.
P264: Wash hands /any other skin areas possibly contacted by the material thoroughly with water after handling.
P273: Avoid release to the environment.

Response:

P305+P351+P338: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310: Immediately call a POISON CENTER, doctor, or for EMERGENCY MEDICAL ASSISTANCE after initiating the first aid measures listed above.
P302+P352: IF ON SKIN: Wash with plenty of water.
P362+P364: Take off contaminated clothing and wash it before reuse.
P321: Specific treatment (see First Aid information on label or Section 4 of the SDS).
P332+P313: If skin irritation occurs: Get medical advice/attention.

P391: Collect spillage.

Disposal:

P501: Dispose of contents/container in accordance with all applicable local, state, and federal regulations. (See Section 13 of the SDS). Be aware that residues in empty containers remain acutely hazardous.

EMERGENCY OVERVIEW

PHYSICAL APPEARANCE: Clear, straw-colored liquid

IMMEDIATE CONCERNS:**OSHA HAZARDS**

Concentrated product causes severe irritation to eyes, and may cause serious injury to eye tissues if not immediately and completely removed. May cause skin irritation, especially with prolonged contact.

COMMENTS: See Section 11: TOXICOLOGICAL INFORMATION.

3. COMPOSITION / INFORMATION ON INGREDIENTS

Chemical Name	Wt. %	CAS
Alkylammonium Methyl Sulfate	12 - 22	proprietary
nonionic surfactant	15 - 25	proprietary
nonionic diluent	< 12	proprietary
Water	Balance	7732-18-5

COMMENTS: Exact percentages of the components of this proprietary formulation are considered confidential. Any additional unlisted components of this proprietary formulation are either considered non-hazardous or are present at concentrations below regulated levels.

4. FIRST AID MEASURES

EYES: Immediately flush eyes with water for at least 15 minutes, forcibly holding eyelids apart to ensure complete irrigation of all eye and eyelid tissues. Remove contact lenses, if present, after the first minute of flushing. "Roll" eyes to fully expose eye surfaces to irrigation. GET IMMEDIATE MEDICAL ATTENTION.

SKIN: Promptly and thoroughly flush contaminated areas with plenty of water for at least 10 minutes. While rinsing, remove all contaminated clothing, jewelry, and shoes immediately, taking care to prevent any contact with the eyes. Thoroughly wash and dry contaminated clothing before re-use. GET PROMPT MEDICAL ATTENTION IF ANY IRRITATION OR ANY OTHER SYMPTOMS DEVELOP.

INGESTION: If swallowed, DO NOT induce vomiting. Never give anything by mouth to a person who is unconscious, convulsing, or unable to swallow. If victim is conscious and able to swallow, immediately rinse out mouth with water, and then give one cup (8 ounces) of milk or water to drink. To prevent aspiration of swallowed material, lay victim on side with head lower than waist. If vomiting occurs spontaneously, keep airway clear. Do not leave victim unattended. GET MEDICAL ATTENTION IMMEDIATELY.

INHALATION: If inhalation exposure to misted material occurs, or if symptoms develop, move to fresh air. If symptoms are severe or persistent, seek prompt medical attention. If breathing is difficult, oxygen should be administered by qualified personnel. If respiration or pulse has stopped, have a trained person administer basic life support (cardiopulmonary resuscitation with rescuer protection and/or Automatic External Defibrillator) as soon as possible, and CALL FOR EMERGENCY MEDICAL SERVICES IMMEDIATELY.

SIGNS AND SYMPTOMS OF OVEREXPOSURE

EYES: May include tearing, stinging, redness, swelling, and blurred vision. Contact with the concentrated product may cause serious, permanent eye damage if not immediately and completely removed.

SKIN: May include drying, defatting, and irritation of skin, progressing to discomfort, itching, chapping, redness, and pain.

INGESTION: May include gagging, coughing, gastrointestinal upset, nausea, vomiting, and diarrhea.

INHALATION: Inhalation of mists and sprays of concentrated product solutions may cause irritation of the respiratory tract, with upper respiratory tract symptoms such as runny nose and sore throat and coughing. Inhalation exposures may aggravate existing respiratory disorders, such as asthma.

NOTES TO PHYSICIAN: Treatment of overexposure should be directed toward the control of symptoms and the clinical condition

of the patient. Keep victim under observation. Symptoms may be delayed.

COMMENTS: Refer to Section 2 (Hazards) and Section 11 (Toxicological Information) for full description of symptoms of overexposure.

5. FIRE FIGHTING MEASURES

FLAMMABLE CLASS: Not combustible unless evaporated to dryness.

GENERAL HAZARD: Spilled material produces extremely slippery surfaces.

EXTINGUISHING MEDIA: Use water fog, carbon dioxide, alcohol-resistant foam, or dry chemical as appropriate to extinguish fire. Avoid using direct water stream on burning liquids due to danger of spreading fire. Use water spray to cool fire-exposed containers.

FIRE FIGHTING PROCEDURES: Material is harmful to aquatic organisms. Prevent fire run-off from entering drains, storm sewers, and natural waterways to prevent damage to the environment. Use water spray to knock down noxious vapors. Evacuate unprotected personnel who are downwind of the fire. Remove containers from fire area if this can be done without risk.

FIRE FIGHTING EQUIPMENT: Wear full turn-out gear and SCBA with full face protection when fighting fires involving industrial chemicals.

HAZARDOUS DECOMPOSITION PRODUCTS: Ignition of evaporated residues produces fumes and smoke, and oxides of carbon, nitrogen, and sulfur.

6. ACCIDENTAL RELEASE MEASURES

SMALL SPILL: Wear appropriate Personal Protective Equipment (as described in Section 8 of the SDS) to prevent any contact with eyes, skin, and clothing. Ensure adequate ventilation. Completely absorb spill with suitable absorbent (e.g., sand, pet litter, universal absorbent, etc.) and scoop up into appropriate sealable containers for proper disposal. Flush remaining trace residues to sanitary sewer with plenty of water.

LARGE SPILL: Wear appropriate personal protective equipment (as described in Section 8 of the SDS) to prevent any contact with eyes, skin, and clothing. Provide adequate ventilation or wear properly fitted respiratory protection equipment. **CAUTION!** Spilled material produces extremely slippery surfaces. Completely contain spilled material with absorbent socks, dikes of sand or earth, etc., to prevent entry into drains, storm sewers, and natural waterways. Consider use of a suitable vacuum or pump to collect as much material as possible in a suitable clean container for use as intended. Use sand, universal absorbent, or similar material to absorb the spill and shovel into closable containers for proper disposal. Remaining trace residues can be mopped and flushed to sanitary sewer with large quantities of water.

ENVIRONMENTAL PRECAUTIONS

WATER SPILL: Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air). Water polluting material. May be harmful to the environment if released in large quantities.

GENERAL PROCEDURES: Keep unnecessary personnel away. Keep people away from and upwind of spill/leak. Wear appropriate protective equipment and clothing during clean-up. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing. Ensure adequate ventilation. Local authorities should be advised if significant spillages cannot be contained. For personal protection, see section 8 of the SDS.

7. HANDLING AND STORAGE

GENERAL PROCEDURES: Always wear the recommended Personal Protective Equipment when handling this material to avoid any contact with eyes, skin, and clothing. (See Section 8.) Provide adequate ventilation. Do not eat or drink or use tobacco products in the work area. Wash hands /any other skin areas possibly contacted by the material thoroughly with water after handling.

HANDLING: Handle and open container with care. Follow all SDS/label precautions even after container is emptied because it may retain hazardous product residues. Clean up spills promptly to prevent slips and falls.

STORAGE: Store in a secured area out of the reach of children. Keep containers tightly closed when not in use. Store away from incompatible materials. Store away from foods, beverages, and feeds.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

EXPOSURE GUIDELINES

OSHA HAZARDOUS COMPONENTS (29 CFR1910.1200)				
	EXPOSURE LIMITS			
Chemical Name	Type		ppm	mg/m ³
Alkylammonium Methyl Sulfate	OSHA PEL	TWA	Not Listed	
		STEL	Not Listed	
	ACGIH TLV	TWA	Not Listed	
		STEL	Not Listed	
nonionic surfactant	OSHA PEL	TWA	Not Listed	
		STEL	Not Listed	
	ACGIH TLV	TWA	Not Listed	
		STEL	Not Listed	
nonionic diluent	OSHA PEL	TWA	Not Listed	
		STEL	Not Listed	
	ACGIH TLV	TWA	Not Listed ^[1]	^[1]
		STEL	Not Listed	
Footnotes: 1. AIHA WEEL TWA = 10 mg/m ³ (ceiling) (AIHA WEEL = American Industrial Hygiene Association Workplace Environmental Exposure Level)				

ENGINEERING CONTROLS: Provide adequate ventilation. Provide local exhaust when mists or aerosols are formed by spray applications. If ventilation is inadequate to maintain airborne concentrations below exposure limits, use properly fitted NIOSH-approved full-face respiratory equipment for protection against mists/ dusts.

PERSONAL PROTECTIVE EQUIPMENT

EYES AND FACE: Wear splash-proof chemical goggles or safety glasses with side shields when handling the concentrated product.

SKIN: Wear appropriate impervious chemical resistant gloves. (Viton, natural rubber, neoprene, nitrile, PVC, polyethylene, and ethyl vinyl alcohol laminate are suitable materials.)

RESPIRATORY: Use of respiratory protection is usually not required. However, if use conditions generate mists, aerosols, or vapors, and ventilation is inadequate, or where symptoms of overexposure are occurring, wear properly fitted NIOSH-approved respiratory equipment for protection against mists and organic vapors.

PROTECTIVE CLOTHING: Wear impervious chemical resistant apron or suit and rubber boots when potential for bodily contact with the concentrated product exists. Contaminated clothing should be immediately removed and laundered.

WORK HYGIENIC PRACTICES: Always use the recommended personal protective equipment to prevent any contact with eyes and skin. Wash skin and contaminated clothing thoroughly after handling. Do not eat, drink or smoke when using this product. Handle in accordance with good industrial hygiene and safety practices.

OTHER USE PRECAUTIONS: A safety shower and eye wash fountain or adequate source of water should be located nearby.

9. PHYSICAL AND CHEMICAL PROPERTIES

PHYSICAL STATE: Liquid

ODOR: Mildly pungent-sharp organic odor

APPEARANCE: Clear, straw-colored liquid

pH: 4.0 to 7.5 (neat liquid)

FLASH POINT AND METHOD: >200°F. Not combustible unless evaporated to dryness.

FLAMMABLE LIMITS: Not determined

AUTOIGNITION TEMPERATURE: Not determined

VAPOR PRESSURE: <24 mm Hg (3.2 kPa) at 77°F / 25°C

VAPOR DENSITY: Not Determined

BOILING POINT: ≥212°F (100°C)

FREEZING POINT: 20° to 22°F / -6.5° to -5.5°C
THERMAL DECOMPOSITION TEMPERATURE: Not Established
SOLUBILITY IN WATER: Complete
PARTITION COEFFICIENT: N-OCTANOL/WATER: Not Established
EVAPORATION RATE: Approximately the same as water.
DENSITY: Approximately the same density as water
SPECIFIC GRAVITY: 1.00 - 1.01
VISCOSITY: Not Established

10. STABILITY AND REACTIVITY

REACTIVITY: No
HAZARDOUS POLYMERIZATION: No
STABILITY: Stable under normal storage and use conditions.
POSSIBILITY OF HAZARDOUS REACTIONS: None known (except for reactions with water-reactive substances and concentrated strong oxidizers).
HAZARDOUS DECOMPOSITION PRODUCTS: Ignition of evaporated residues produces fumes and smoke, and oxides of carbon, nitrogen, and sulfur.
INCOMPATIBLE MATERIALS: Concentrated strong oxidizing agents. Water reactive materials.

11. TOXICOLOGICAL INFORMATION

ACUTE TOXICITY

DERMAL LD₅₀: No testing data are available for the product as a whole. Based on the weighted values of the reported dermal toxicities of the components, the estimated acute dermal LD₅₀ is greater than 4000 mg/kg.

ORAL LD₅₀: No quantitative testing data are available for the product as a whole. The estimated LD₅₀ value (rat) for the product is greater than 5,000 mg/kg, based on the weighted average of the LD₅₀ values for the individual components.

INHALATION LC₅₀: No quantitative testing data are available for the product as a whole. Prolonged or chronic Inhalation of mists or sprays of concentrated solutions of the product may be harmful. Do not breathe spray mists of the undiluted product. Harmful effects of inhalation will depend upon solution strength and duration of exposure.

RESPIRATORY OR SKIN SENSITIZATION: No reports of sensitization have been found in the health and safety literature for any of the individual components present at ≥ 0.1%.

GERM CELL MUTAGENICITY: No reports of mutagenicity have been found in the health and safety literature for any of the components present at ≥ 0.1%.

CARCINOGENICITY

IARC: No component of this product present at levels greater than or equal to 0.1% is listed as a probable or confirmed human carcinogen by the International Agency for Research on Cancer (IARC).

May contain trace levels of Oxirane (CAS 75-21-8), 1,4-Dioxane (CAS 123-91-1), and Acetaldehyde (CAS 75-07-0) that are listed as known or suspected carcinogens by the IARC and other regulatory agencies.

NTP: No component of this product present at levels greater than or equal to 0.1% is listed as a probable or confirmed human carcinogen by the National Toxicology Program.

OSHA: No component of this product present at levels greater than or equal to 0.1% is listed as a probable or confirmed human carcinogen by OSHA.

REPRODUCTIVE TOXICITY: None reported from laboratory studies on the individual components at doses below maternally toxic levels.

STOT-SINGLE EXPOSURE: Not classified as a specific organ toxicant for single exposure or repeated exposure.

GENERAL COMMENTS: No data for other effects are available.

12. ECOLOGICAL INFORMATION

ECOTOXICOLOGICAL INFORMATION: This material exhibits significant acute and chronic toxicity to aquatic organisms. Do not contaminate natural waters by spills or discharges.

BIOACCUMULATION/ACCUMULATION: No bioaccumulation data are available for the product.

AQUATIC TOXICITY (ACUTE): No testing data are available for the product as a whole. Acute LC₅₀ values for freshwater fish (96 hr) and EC₅₀ values for daphnia (48 hr) are estimated to be in the range of 1-30 mg/L, based on weighted averages of literature toxicity values for the individual components.

Notes: Toxicity effects in the aquatic environment are attributable to the surfactant properties of the detergent blend.

GENERAL COMMENTS: Do not dump on the ground, into storm sewers and drains to natural waters, or any body of water. Large

spills or discharges that may adversely affect the environment should be reported immediately to federal, state and local authorities.

13. DISPOSAL CONSIDERATIONS

PRODUCT DISPOSAL: Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and all applicable federal, state, and local regulations. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and storm sewers to prevent harm to the environment.

EMPTY CONTAINER: Be aware that product residues in empty containers remain acutely hazardous. Follow all SDS and label precautions even after container is emptied because of the hazards posed by product residues.

RCRA/EPA WASTE INFORMATION: As sold, product is classified by RCRA regulations as non-hazardous for purposes of disposal.

14. TRANSPORT INFORMATION

DOT (DEPARTMENT OF TRANSPORTATION)

PROPER SHIPPING NAME: UN3082, Environmentally Hazardous Substances, Liquid, N.O.S. (contains Alkylammonium Methyl Sulfate), 9, PGIII

REPORTABLE QUANTITY (RQ) UNDER CERCLA: Not applicable to non-bulk packages (less than 5000 gallons) of this product.

MARINE POLLUTANTS: Alkylammonium Methyl Sulfate

15. REGULATORY INFORMATION

UNITED STATES

SARA TITLE III (SUPERFUND AMENDMENTS AND REAUTHORIZATION ACT)

FIRE: No **PRESSURE GENERATING:** No **REACTIVITY:** No **ACUTE:** Yes **CHRONIC:** No

313 REPORTABLE INGREDIENTS: Not Applicable. (No listed chemicals are present at or above the *de minimis* levels that require reporting under Section 313.)

302/304 EMERGENCY PLANNING

THRESHOLD PLANNING QUANTITY / REPORTABLE QUANTITIES: No components on the Section 302 or Section 304 Extremely Hazardous Substances lists are present at concentrations above threshold levels.

CERCLA (COMPREHENSIVE RESPONSE, COMPENSATION, AND LIABILITY ACT)

CERCLA RQ: No CERCLA listed components are present at levels that would require reporting for spills/releases of less than 10,000 gallons of product.

TSCA (TOXIC SUBSTANCE CONTROL ACT)

TSCA STATUS: All components of this product are listed on the Toxic Substance Control Act (TSCA) inventory, or are exempt.

16. OTHER INFORMATION

REASON FOR ISSUE: New SDS format (GHS compliant). This version replaces all previous versions.

APPROVED BY: Technical Director

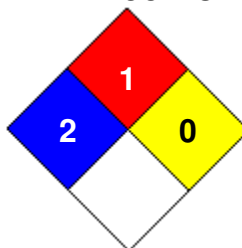
PREPARED BY: Technical Services - Broadmoor Products **Date Prepared:** 02/01/2016

INFORMATION CONTACT: Technical Services - Broadmoor Products, Inc.: (616) 285-6440

HMIS RATING

HEALTH	2
FLAMMABILITY	1
PHYSICAL HAZARD	0
PERSONAL PROTECTION	C

NFPA CODES



GENERAL STATEMENTS: Each user of this product is expected and urged to read and understand the entire SDS (consulting appropriate expertise as necessary) in order to comprehend the information contained in this SDS and any hazards associated with the product, with the expectation that the precautions identified in this document will be followed unless your use conditions

would necessitate other appropriate methods or actions.

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